

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 8

Linear Regression Review

Instructor: Yan Li

e-mail: yli6@umd.edu

Discussion Outline

PART 1

- Simple Linear Regression
- Ordinary Least Squares (OLS)
- Model Building
- Regression Diagnostics
- Transformations
- Weighted Least Squares (WLS)

PART 2

- Linear Regression for Complex Sample Survey Data

Regression Analysis

- **Regression Analysis:** A study of the relationship between variables.
- Two classes of variables: Response or dependent variables (“left hand side variables”) and predictor or independent variables (“right hand side variables,” covariates, explanatory variables, regressors, etc.).
- Objective: To *infer* about the expected value of the response or some characteristic of the distribution of the response variable *conditional* on the predictors.
- Predictors are assumed to be known or fixed constants.
- **Linear Model:** Refers to a class of models used to define this relationship that are linear functions of unknown parameters, β .
- y = Response variable, x = predictor variables

Steps in a Regression Analysis

- 1. Postulate a model.** General behavior of the responses on the subjects for a given set of values of the predictors. The model is based on subject matter knowledge and empirical investigation of the data. The model will have a certain number of unknown quantities called parameters. Not all subjects will exactly follow the pattern specified by the model. The statistical errors or residuals are simple devices to account for the model's failure to provide an exact fit.
- 2. Analyze data to obtain the estimates of the parameters and their standard errors.**
- 3. Diagnostics are used to examine the suitability of model.** This may lead to a modification of the model structure. We will then cycle back to the model fitting and model diagnosis process.
- 4. The estimates and standard errors from the final model are then used characterize or infer about the conditional distribution of y given x .**

Structure of the Regression Model

- Regression models the relationship of a function of the expected value of the dependent variable, $g(E(y))$, to the independent variables $\mathbf{x} = \{x_1, \dots, x_p\}$.

For regression models that are linear in the parameters:

$$g(E(y)) = g(\hat{Y}) = \beta_o + \beta_1 x_1 + \dots + \beta_p x_p$$

Structure of the Regression Model

- The function $g(\cdot)$ is called the link function
- Common links in applied regression analysis
 - Linear regression: Identity, $g(E(y))=E(y)$
 - Logistic regression: Logit, $g(E(y))=\log[E(y)/(1-E(y))]$
 - Probit regression: Probit, $g(E(y))=\Phi^{-1}(E(y))$
 - Poisson, NB: $g(E(y))=\log(E(y/t))=\log(E(y))-\log(t)$

Simple Linear Regression

Our focus will be on a broad class of models (at least for now) called ***Linear Models***. ***These models are linear functions of unknown parameters.***

- Linear models

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$$

$$\mathbf{y} = \mathbf{\beta x} + \epsilon$$

- Nonlinear models

$$y = \beta_0 + \beta_1 x^{\beta_2} + \epsilon$$

$\beta_0, \beta_1, \beta_2$: unknown parameters

Estimation of the Parameters

- Approach to parameter estimation will depend on the link function, $g(\cdot)$ and the assumed distribution of Y
- For example:
 - Linear regression
 - $g(E(y))=E(y)$, $Y \sim N(X\beta, \sigma_{y,x})$
 - Logistic
 - $g(E(y))=\text{logit}(E(y))$, $Y \sim B(1, \pi(X))$

Simple Linear Regression Model

M1: $y_i = x_i\beta + \epsilon_i$ (*vector notation*)
 $= \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \epsilon_i,$

where

$$y_i \sim \text{Normal}(x_i\beta, \sigma^2)$$

$$\beta = \{\beta_0, \beta_1, \dots, \beta_p\}'$$

ϵ_i = residual for subject i

$$E(\epsilon_i | x_i) = 0$$

$$\text{Var}(\epsilon_i | x_i) = \sigma^2 (\text{constant variance})$$

$$\text{Cov}(\epsilon_i, \epsilon_j | x_i, x_j) = 0 (\text{uncorrelated})$$

Simple Linear Regression Model

Implications:

$$E(y_i|x_i) = E(x_i\beta) + E(\epsilon_i) = x_i\beta + 0$$

$$= x_i\beta \text{ (regression function)}$$

$$= \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi},$$

with

$$x_i = \{1, x_{1i}, \dots, x_{pi}\} \text{ and}$$

$$\beta = \{\beta_0, \beta_1, \dots, \beta_p\}'$$

$$\epsilon_i|x_i \sim \text{Normal}(0, \sigma^2)$$

$$\text{Var}(y_i|x_i) = \sigma^2$$

$$\text{Cov}(y_i, y_j|x_i, x_j) = 0$$

Interpretation of Regression Coefficients

- Simple Linear regression model

$$E(y|x = x_0) = \beta_0 + \beta_1 x_0$$

$$E(y|x = x_0 + 1) = \beta_0 + \beta_1 x_0 + \beta_1$$

- On the average, the expected change in y for unit change in x

$$\beta_1 = E(y|x = x_0 + 1) - E(y|x = x_0)$$

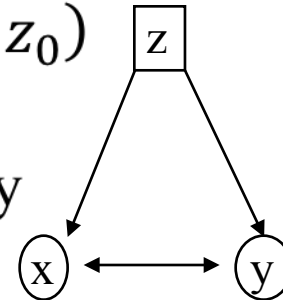
- Add a confounding variable

$$E(y|x = x_0, z_0) = \beta_0 + \beta_1 x_0 + \beta_2 z_0$$

$$E(y|x = x_0 + 1, z_0) = \beta_0 + \beta_1 x_0 + \beta_1 + \beta_2 z_0$$

$$\beta_1 = E(y|x = x_0 + 1, z_0) - E(y|x = x_0, z_0)$$

β_1 = the “expected” (average) change in y for one unit increase in the independent variable x , holding z constant.



A portion of x - y relationship is attributable to z

Interpreting Linear Regression Coefficients: Intercept Terms

- Intercept $\beta_0 = E(y|x = 0, z = 0)$ is often not of much interest as it may represent expectation outside the range of data
- Re-formulation (centering) of the predictors in the model can make intercept interpretable

$$y = \beta_0^* + \beta_1(x - \bar{x}) + \beta_2(z - \bar{z}) + \epsilon,$$
$$\beta_0^* = \bar{y} = \textit{“Centercept”}$$

- Models with no intercept force the regression through $y=0$, $x=0$.

Interpreting Linear Regression Coefficients: Nonlinear and Interaction terms

- Caution about interpreting with nonlinear terms:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon,$$

β_1, β_2 linked

β_1 alone in the presence of β_2 has no interpretation

β_2 measures the extent of nonlinearity

- Or interaction terms:

$$y = \beta_0 + \beta_1 x + \beta_2 z + \gamma_1 xz + \epsilon,$$

$\beta_1, \beta_2, \gamma_1$ linked

To determine the effect of an increase in x on the expected value of y , must evaluate the functions at x and $(x+1)$ and take the difference.

Indicator Variables

(a.k.a. dummy variables)

- Required to represent effects of levels of categorical variables in regression models
- Several ways to specify, which determines how coefficients are interpreted
- Here: $(0,1)$ “parameterization”

Indicator Variables: Reference Category

e.g. $X = \text{Race} = \begin{matrix} 1 & \text{White} \\ 2 & \text{Black} \\ 3 & \text{Other} \end{matrix}$

Choose 1 reference category:

- Choose to aid your research objective
- **Some choose most prevalent group**

Indicator Variables: Category Examples

Race

1	2	3
---	---	---

Region

1	2	3	4
---	---	---	---

Cataract

1	2
---	---

Indicator Variables: Example (0,1 Method)

$$\text{Model: } Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

$$X_1 = \begin{cases} 1 & \text{if race} = 2 \\ 0 & \text{otherwise} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{if race} = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Black R: } Y = \beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 0$$

$$\text{Other R: } Y = \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1$$

$$\text{White R: } Y = \beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 0$$

Indicator Variables: Interpreting coefficients

β_1 is the effect on $E(g(y))$ of being Black vs. White

β_2 is the effect of being Other vs. White

β_0 is the effect of being White on $E(g(y))$

For this reference parameterization,
the effect of being Black vs. Other is:

$$\begin{aligned} E\left(Y_{(Black)} \mid \underline{x} \right) - E\left(Y_{(Other)} \mid \underline{x} \right) \\ = \left[\beta_1 \cdot 1 \right] - \left[\beta_2 \cdot 1 \right] \\ = \beta_1 - \beta_2 \end{aligned}$$

?

Software Defaults

- ✓ Stata: **svy: regress** command
 - Indicate categorical predictors in the list of predictors with `i.catvar`
 - Use `ib3.catvar`, for example, to set 3 as the reference category
 - This syntax will be demonstrated in examples

Estimation Method:

Least Squares Estimation

- By far the most popular method for estimating the unknown parameters is the Method of Least Squares.
- Estimate β by minimizing the residual sum of squares,

$$\sum_{i=1}^n (y_i - x_i\beta)^2$$

- Estimate of σ^2 :

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - x_i\beta)^2}{n - p - 1}$$

p = number of covariates in the regression Model of M1

Properties of Least Squares Estimate

- Analytically simple. It is a linear function of the response values.
- Unbiased: $E(\hat{\beta}|x) = \beta$
- It has the lowest variance among all other unbiased estimates that are also linear functions of the response values.
(B.L.U.E.)
- Under normal errors, the least squares estimates are also maximum likelihood estimates.
- Sampling distribution of the regression coefficients follows a t -distribution

$$\frac{\hat{\beta}_k - \beta_k}{se(\hat{\beta}_k)} \sim t_{n-p-1}$$

- It is sensitive to outliers, influential points, heteroskedastic variances!

Least Squares Regression: Comparison of Two Models

$$x = (x_1, x_2)$$

$$x_1 = p_1 \text{ predictors}$$

$$x_2 = p_2 \text{ predictors}$$

Compare:

$$y = x\beta + \epsilon \text{ (Full model)}$$

vs.

$$y = x_1\beta_1 + \epsilon \text{ (reduced model)}$$

Under $H_0: \beta_2 = \{\beta_{p_1+1}, \dots, \beta_p\} = \{0, \dots, 0\}$

$$F = \frac{\frac{SSE(reduced) - SSE(full)}{(n - p_1 - 1) - (n - p_1 - p_2 - 1)}}{\frac{SSE(full)}{(n - p_1 - p_2 - 1)}} \sim F(p_2, n - p_1 - p_2 - 1)$$

Confidence Interval (CI) for $E(y|x)$

Prediction Interval Unknown y Given x

x_f : Predictors; y_f : Unobserved y

$$\underline{\hat{E}(y_f|x_f)} = x_f \underline{\hat{\beta}}$$

$$\text{var}(\hat{E}(y_f|x_f)) = x_f \Sigma(\hat{\beta}) x_f$$

$$100(1-\alpha)\% \text{ CI: } x_f \hat{\beta} \mp t_{1-\frac{\alpha}{2}, n-p-1} \sqrt{\text{var}(\hat{E}(y_f|x_f))}$$

$$\underline{\hat{y}_f} = x_f \underline{\hat{\beta}} + \epsilon$$

$$\text{var}(\hat{y}_f) = x_f \Sigma(\hat{\beta}) x_f + \hat{\sigma}^2$$

$$100(1-\alpha)\% \text{ prediction limits: } x_f \hat{\beta} \mp t_{1-\frac{\alpha}{2}, n-p-1} \sqrt{\text{var}(\hat{E}(y_f|x_f)) + \hat{\sigma}^2}$$

Diagnostics, Model Fit

- Non-normality of residuals
 - Diagnostic Plots (x-y, Q-Q plots, histograms)
 - Tests (e.g. Kolmogorov-Smirnov; very powerful against H_0)
- Heterogeneity of Variance
 - Diagnostic Plots
- Outliers
 - Residual plots
 - Influence statistics
 - Studentized residuals (y)
 - Hat statistics (leverage)
 - Cook's D (influence)
 - Neter et al. (1996) for details

Residual Plots

- * The residuals from the fitted model,

$$e_i = (y_i - \hat{y}_i) = (y_i - x_i\hat{\beta}); i = 1, \dots, n$$

provide information about the appropriateness of the model.

- * Most diagnostic procedures involve inspecting the plots of the residuals.
- * A plot of the residuals against the predictors can indicate non-linearity, non-constant variance (here a plot against the predicted values may be useful).
- * Histogram of the residuals will be useful to detect departures from normality.
- * A quantile-quantile plot is another graphical tool useful for detecting non-normality (this is a plot of quantiles estimated from residuals against those computed from a normal or theoretical distribution; a straight line indicates normality).

Transformation to Normality

- A common remedial measure for non-normality or heterogeneity of variance is transformation of the dependent variable.
- One has to be careful because it destroys the nice interpretation of the parameters.
- A common transformation that is often used (and still has nice interpretation) is the natural (base e) log-transformation:

$$\text{Log } y = \beta_0 + \beta_1 x + \epsilon$$

$$\frac{E(y|x = x_0 + 1)}{E(y|x = x_0)} = e^{\beta_1}$$

Weighted Least Squares

- When the residual plots indicate evidence of non-constant variance, either the dependent variable is transformed or the variance can be directly modeled.
- When the variance is modeled, a **weighted least squares (WLS)** approach is used to estimate the parameters.
- Weights in WLS are inversely proportional to the residual variance.

$$WRSS = \sum_i^n w_{i,ls} (y_i - x_i \hat{\beta})^2,$$
$$x_i = (1, x_{i1}, \dots, x_{ip})$$

Example: heterogeneity of variance that is a function of a predictor variable.

$$\text{var}(e_i) = \sigma^2 x_{i1}$$

$$x_{i1} = \text{predictor (positive)}$$

$$W_i = \frac{1}{\sigma^2 x_{i1}}$$

$$WRSS = \sum_i^n \frac{(y_i - x_i \hat{\beta})^2}{\sigma^2 x_{i1}},$$

where

$$x_i = (1, x_{i1}, \dots, x_{ip})$$